

RAHMAN NENDHIARTO

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AI & Machine Learning specialist with expertise in Software Development, experienced in building intelligent digital products through modern software engineering and data-driven technologies. Developed predictive AI models, end-to-end web applications, and interactive analytics platforms from concept to deployment through research, client projects, and professional internships. Passionate about delivering scalable, maintainable, and user-centered software that transforms complex data into impactful real-world solutions.

EDUCATION

Universitas Ahmad Dahlan

Yogyakarta, Indonesia

Bachelor of Informatics Engineering | GPA: 3.52/4.00

2022 - 2026

Relevant Coursework: Machine Learning, Artificial Intelligence, Software Engineering, Database Systems, Web Programming.

Thesis: PM2.5 Air Quality Prediction Using a Hybrid Sine Cosine Algorithm and Long Short-Term Memory.

EXPERIENCE

Universitas Ahmad Dahlan

Yogyakarta, Indonesia

Machine Learning Developer

2025 - Present

Developed an end-to-end PM2.5 air quality forecasting solution by integrating the Sine Cosine Algorithm (SCA) with Long Short-Term Memory (LSTM) to optimize predictive performance. Engineered the complete machine learning pipeline, including data preprocessing, feature engineering, hyperparameter tuning, and performance evaluation using RMSE, MAPE, and R^2 . Designed and deployed UdaraCast, a web-based forecasting platform that enables users to explore interactive PM2.5 predictions and environmental insights across multiple cities and years.

Ministry of Industry of Indonesia

Yogyakarta, Indonesia

Full Stack Developer Intern

2025

Developed and maintained SIMAGANG, a web-based internship registration and management platform that digitizes internship administration at BBSPJIKB Yogyakarta. Implemented frontend and backend features, authentication, database integration, and workflow automation using Laravel, MySQL, and modern web technologies. Collaborated with cross-functional teams to deliver a secure, scalable, and user-centered application that improved internship registration and management processes.

Istana Sayur Buah

Manokwari, Indonesia

Lead Developer & Project Manager

2025 - Present

Led the end-to-end development of a production-ready e-commerce platform for a fresh produce retailer, from business requirement analysis through deployment. Directed project planning, coordinated development activities, and implemented a responsive full-stack application featuring product catalog management, shopping workflows, and customer-focused user experiences. Delivered a scalable digital solution while strengthening leadership, stakeholder communication, and project management capabilities.

Digital Talent Scholarship - Ministry of Communication and Digital Affairs (Komdigi)

Yogyakarta, Indonesia

AI & Machine Learning Trainee

Aug 2025

Completed an intensive project-based training program in Artificial Intelligence, Machine Learning, and Data Science. Developed end-to-end machine learning workflows involving data preprocessing, feature engineering, predictive modeling, and model evaluation using Python, Pandas, Scikit-learn, and TensorFlow. Applied software engineering

principles to build AI-driven solutions for real-world datasets while strengthening practical skills in intelligent application development.

National Professional Certification Board (BNSP) - Ahmad Dahlan Professional Certification Institute
Associate Data Scientist Candidate

Yogyakarta, Indonesia
Sep 2025

Successfully completed a national competency assessment for the Associate Data Scientist qualification, demonstrating practical expertise in Artificial Intelligence, machine learning, predictive analytics, data preprocessing, feature engineering, and model evaluation. Validated technical competencies through competency-based assessments aligned with the Indonesian National Work Competency Standards (SKKNI).

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, PHP, HTML5, CSS3

Artificial Intelligence & Machine Learning: TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Software Development: Laravel, React.js, Next.js, Node.js, CodeIgniter, Tailwind CSS, Bootstrap

Databases & Deployment: MySQL, Supabase, Vercel

Data Visualization: Power BI, Tableau

Developer Tools: Git, GitHub, Google Colab, Cursor, Figma, XAMPP

PROJECTS

Global PM2.5 Air Quality Forecasting Platform - UdaraCast
Undergraduate Thesis Project

Yogyakarta, Indonesia
Jan 2025 – Present

Developed an AI-powered air quality forecasting platform by integrating the Sine Cosine Algorithm (SCA) with Long Short-Term Memory (LSTM) using Python, TensorFlow, Scikit-learn, Pandas, Laravel, and MySQL. Engineered an end-to-end machine learning pipeline, including data preprocessing, feature engineering, hyperparameter optimization, and model evaluation using RMSE, MAPE, and R^2 . Deployed the predictive model as an interactive web application, enabling users to explore air quality forecasts and environmental insights across multiple cities. Achieved an RMSE of 0.1042, MAPE of 0.8847, and an R^2 score of 0.9787, demonstrating high predictive accuracy and model reliability.

Exporetrade - Global Product Sourcing Platform
Client Project

Yogyakarta, Indonesia
Jun 2025 – Present

Designed and developed a scalable global product sourcing platform using Next.js, React.js, JavaScript, Tailwind CSS, and Supabase, connecting international buyers and suppliers through product requests, quotation management, and supplier discovery. Built responsive user interfaces, integrated cloud-based databases, and optimized application architecture to streamline business collaboration and support digital export promotion.

SIMAGANG - Internship Management Information System
Internship Project

Yogyakarta, Indonesia
Des 2025 – Jan 2026

Developed a web-based internship management system for BBSPJIKB Yogyakarta, Ministry of Industry of Indonesia using Laravel, MySQL, JavaScript, and Bootstrap. Implemented secure authentication, database integration, and responsive user interfaces to digitize internship registration, participant management, and administrative workflows, improving operational efficiency while reducing manual administrative processes.

Pipe Thickness Loss Prediction Dashboard
Machine Learning Project

Yogyakarta, Indonesia
Apr 2025 - Present

Built an interactive predictive maintenance dashboard using Python, TensorFlow, Plotly, and Power BI to forecast industrial pipeline thickness loss with Long Short-Term Memory (LSTM). Developed the complete machine learning workflow, including data preprocessing, model training, performance evaluation, and interactive visualization to support engineering analysis and data-driven maintenance planning. Achieved an R^2 score of 0.9795, demonstrating strong predictive performance for industrial maintenance forecasting.

Led the end-to-end development of a production-ready e-commerce platform using Laravel, MySQL, JavaScript, and Bootstrap for a fresh produce retailer. Designed and implemented product catalog management, shopping workflows, secure database integration, and responsive user interfaces, delivering a scalable and user-centered online shopping experience.